A FINITE-RANK HANKEL FORM MISSED BY POINT JETS

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> For the one complex dimensional Riemann surface of Example 3.4 it seems reasonable to conjecture that every finite rank symbol is of the form $b(\cdot) = \sum_{k=1}^{N} a_k Z^{n_k} K_{z_k}(\cdot)$ for some constants a_k, z_k, n_k.

Deck and Gross, *Hankel forms on complex manifolds*, doi:10.2140/pjm.2002.205.43, Remark 6.10, p. 94.

Example 3.4 is the n-sheeted cover of $\mathbb{C} \setminus \{0\}$ for $z^{1/n}$, with the lifted Gaussian measure. Its spectral algebra is spanned by $z^{k/n}$, $k \geq 0$. Here K_{z_k} is the reproducing kernel at z_k, and Z is the lift of the Gaussian Euler vector field $a^{-1} z \partial_z$ in the covering coordinate; $Z^{n_k} K_{z_k}$ represents a derivative evaluation.

Answered: no. Evidence: Sketch.

No. On the literal punctured surface, the constant symbol gives a bounded rank-one Hankel form that is not a finite sum of the proposed point jets.

Proof. On the Riemann surface of $z^{1/n}$, the coordinate $w = z^{1/n}$ identifies the surface with $\mathbb{C} \setminus \{0\}$, while its spectral algebra is $\mathbb{C}[w]$. For the lifted Gaussian measure with parameter $a > 0$, angular integration gives

$$\|w^k\|^2 = (2a)^{k/n} \Gamma(1 + k/n), \qquad k \geq 0.$$

The monomials are orthogonal, so evaluation at the omitted point is bounded: $|P(0)| = |\langle P, 1 \rangle| \leq \|P\|$. Therefore $\Gamma_1(P, Q) = \langle PQ, 1 \rangle = P(0)Q(0)$ is a bounded rank-one Hankel form.

A finite sum of point jets at nonzero points $w_1, \ldots, w_s$ annihilates

$$P(w) = \prod_{j=1}^{s} (1 - w/w_j)^{m_j + 1},$$

where m_j bounds the jet order at w_j. Yet $P(0) = 1$, so the functional $P \mapsto P(0)$ does not lie in the span of these jets. This refutes the literal point-support conjecture; it does not classify finite-rank symbols on other analytic spaces. □

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